

Unit - 1

Introduction

Errors in Numerical Computations — When we go for numerical computation, as a rule, we deal with approximate values of data (with approximate number). The initial data themselves are usually given with some errors, which are then increased in the course of calculation by rounding errors, the use of numerical (approximate) formulae, and so on. A reasonable error estimation empowers us to decide on the number of digits to be retained in intermediate operations, and in the final result. The quantity, True data - Approximate data is called error.

In order to determine the accuracy in an approximate solution to a problem, either we find the bound of the

$$\boxed{\text{Relative error} = \frac{|\text{Error}|}{|\text{True data}|}} \quad \text{————— ①}$$

For the estimation of error of an approximate number x means to establish an inequality of the form

$$\boxed{|x - x_0| \leq \epsilon_x} \quad \text{————— ②}$$

The number ϵ_x is called the absolute error of an approximate number x .

Accuracy of Numbers —

- (a) Approximate number — We can classify the numbers in two ways — exact and approximate. Exact means which has a finite number of digits after decimal, otherwise they are said to be approximate.
eg. Sometimes we write $\frac{1}{3} = 0.33$, $\sqrt{2} = 1.414$ & $\pi = 3.141$, but 0.33 is not exact value of $\frac{1}{3}$, it is an approximation.

(b) Significant figures:- Numbers 1254 consists 4 digits, 3.141 also consists 4 digits while 0.0221 consists only 3² digits, so we can call it as number 1254 & 3.141 contains four significant figures while the number 0.0221 contains three significant figures. Hence, the significant figures are the number of digits used to express a number.

(c) Rounding off:- Sometimes we have the large numbers like 1.333... or 3.142857.... In practice, it is very typical to consider all the figures of these numbers. So we try to restrict it upto a certain significant figure like 1.33 or 3.142. This process of dropping unwanted digits is called rounding off.

According to standard practice, we use here the following rounding off rule (rule) -

"If the first discarded digit exceeds or equals 5, add 1 to this last retained digit otherwise 0." e.g.

$$1.325 \approx 1.33 \rightarrow e_a = 0.5 \times 10^{-2}$$

$$1.324 \approx 1.32 \rightarrow e_a = 0.4 \times 10^{-2} < 0.5 \times 10^{-2}$$

$$3.674 \approx 3.67 \rightarrow$$

$$19.656 \approx 19.66$$

$$5.8257 \approx 5.825$$

In mathematical tables, all the numbers are rounded to correct digits, i.e., all indicated significant digits are correct, and the absolute error does not exceed half the unit of the last digit retained, e.g., if in a table $\pi = 3.142$, then the absolute error does not exceed 0.5×10^{-3} .^④

Errors in Numbers — Following types of errors are encountered in performing numerical calculations —

- ① Inherent errors — The error which is already exists in the given data of the problem, called inherent error. It can be sort out by taking better data by using high precision computing aids. (It is generally because of approximation of the data).
- ② Errors due to finite representation — e.g. representation of $\frac{1}{3}$ using finite no.
- *③ Rounding errors — Errors due to arithmetic operations using normalized floating point numbers. The rounding error is the quantity which must be added to the finite representation of a computed number in order to make it the true representation of that number. It can be reduced by changing the calculation procedure so as to avoid subtraction of nearly equal numbers or division by smaller number, or by retaining at least one more significant figure at each step than that given in the data and rounding off at the last step.
- ④ Truncation error : — Errors due to finite representation of an inherently infinite process. e.g. due to the use of finite terms in the infinite series expansion of $\sin x$, $\cos x$ etc.

One of these, the rounding error is the most difficult to estimate and one has to follow a set of thumb rules to reduce its effect on the final results. There are two measures commonly used. These are —

- ① A measure of absolute error. (E_a)
 - ② A measure of relative error. (E_r)
- ① Measure of Absolute Error (E_a) — If m_1 is the correct value of a variable and m_2 the value obtained by computation then $|m_1 - m_2| = E_a$ is called the absolute error. If the value of m_1 is near unity then it is a good measure of error.

* Q — How rounding errors can be reduced?

$\Rightarrow$ If $m_1 = 10^{-6}$ & $m_2 = 0.5 \times 10^{-6}$ then $e_a = 0.5 \times 10^{-6}$ which seems very small but the error is considerable.

$\Rightarrow$ If $m_1 = 10^6$ & $m_2 = 0.99 \times 10^6$ then $e_a = 10^4$ which seems very large but we know that the error in this case should be smaller than that of in the previous case. In such cases absolute error has little meaning and one has to have a measure of relative error.

② Measure of Relative Error (e_r) — A measure of relative error is defined as

$$\left| \frac{m_1 - m_2}{m_1} \right| = \left| \frac{e_a}{m_1} \right| = e_r$$

Using this defⁿ in first example, we have

$$e_r = \left| \frac{10^{-6} - 0.5 \times 10^{-6}}{10^{-6}} \right| = 0.5$$

& for second example

$$e_r = \left| \frac{10^6 - 0.99 \times 10^6}{10^6} \right| = 0.01$$

Now, the relative error in the second case is $(1/50)^{th}$ of that in the first case.

Significant Digits: — When this term is used to describe numbers then it refers to the number of digits in a number excluding leading zeros.

For instance the number 158, 1.58, 1.08, 0.00108 all have three significant digits. When this term is used to describe results obtained from calculations then it is used to signify the number of digits in the answer (excluding leading zeros) which are known to be correct.

For instance, if we say that $x = 3.452$ is correct to 4 significant digits then we mean that

$$(3.452 - 0.0005) \leq x \leq (3.452 + 0.0005)$$

The relative error

$$\epsilon = \frac{0.001}{3.452} = 0.00029 = 0.29 \times 10^{-3}$$

i.e. $3.4515 \leq x \leq 3.4525$
 $\begin{array}{r} 3.4525 \\ - 3.4515 \\ \hline 0.001 \end{array}$

In trial & error method of solving problem (i.e. iterative method) using normalized floating point computations we are required to stop the trial & error cycle when the required significant digits in the answer are assured.

* Let us see what maximum error in the result may be allowed to guarantee a certain number of significant digits. Let us consider a number 0.1000 which is good to 4 significant digits. Then

$$0.09995 \leq 0.1000 \leq 0.10005$$

Relative error,

$$\epsilon = \frac{0.0001}{0.1} = 0.001$$

For the number .9999 which is also good to 4 significant digits and

the relative error,

$$\epsilon = \frac{0.0001}{0.9999} = 0.00010001 > 0.001$$

The relative error for a 4 digit normalized floating point number thus lies between 0.0001 & 0.001. In order to guarantee 4 significant digit accuracy we should ensure that the relative error is $\leq 0.0001 = 10^{-4}$. In general to guarantee n significant digit accuracy we should ensure that $|\epsilon| \leq 10^{-n}$.

Percentage Error (C_p) —

$$C_p = C_r \times 100$$

Q- Show that in order to guarantee n significant digit accuracy for a normalized floating point number should be $\leq 10^{-n}$.

Q1: Round off the following numbers to four significant digits and then calculate absolute error, relative error & percentage error.

(i) 75462 (ii) 537.261 (iii) 21.46235

Solⁿ: (i) Number is 75462 (= n say)

After rounded off to four significant figures = 75460 (= n₁ say)

Now, absolute error, $E_a = |n - n_1| = |75462 - 75460| = 2$.

$$\text{relative error, } e_r = \frac{|E_a|}{n} = \frac{2}{75462} = 2.65 \times 10^{-5}$$

$$\text{percentage error, } e_p = e_r \times 100 = 2.65 \times 10^{-5} \times 100 = 2.65 \times 10^{-3}$$

(ii) Number is 537.261 (= n say)

After rounded off to four significant figures = 537.3 (= n₁ say)

Now, absolute error, $E_a = |n - n_1| = |537.261 - 537.300| = 0.039$

$$\text{relative error, } e_r = \left| \frac{n - n_1}{n} \right| = \frac{0.039}{537.261} = 7.25 \times 10^{-5}$$

$$\text{percentage error, } e_p = e_r \times 100 = 7.25 \times 10^{-5} \times 100 = 7.25 \times 10^{-3}$$

Note $\Rightarrow$ If the first discarded digit exceeds or equals 5 then add 1 to last retained digit otherwise 0.

We have the following rules regarding the absolute and relative errors -
if $x = 0.d_1d_2d_3 \dots \times 10^n$, where $d_1, d_2, d_3, \dots$ are decimal digits then

1. Absolute Error because of Truncation - If x_a is the approximate value (i.e. approximation) of x after truncation to m digits, then

$$|x - x_a| < 10^{n-m}$$

② Relative Error because of Truncation - If x_a is the approximation of x after truncation to m digits then $\left| \frac{x - x_a}{x} \right| < 10^{-m+1}$

③ Absolute Error because of Rounding off - If x_a is the approximation of x after Rounding off to m digits, then

$$|x - x_a| < 0.5 \times 10^{n-m}$$

④ Relative Error because of Rounding off - If x_a is the approximation of x after Rounding off to m digits, then

$$\left| \frac{x - x_a}{x} \right| < 0.5 \times 10^{-m+1}$$

(Numeric Computation)

Computer Arithmetic - Whenever we go for computation with digital computers, the first step is to convert the decimal number into the another number system with an appropriate base, so that, a computer can understand it and then after it can store these converted numbers in its memory. The memory of the digital computer is divided into separate cells called words, and each word can hold the same number of digits called bits, with respect to its base plus a sign. The number of digits, a computer word can store is called word length, which can varies from one computer to another. The numbers in computer word can be stored in two ways -

- (i) Fixed point
- (ii) Floating point

Floating Point Representation of Numbers - There are two types of arithmetic operations available

in a computer. These are:

- (i) Integer arithmetic
- (ii) Real or floating point arithmetic.

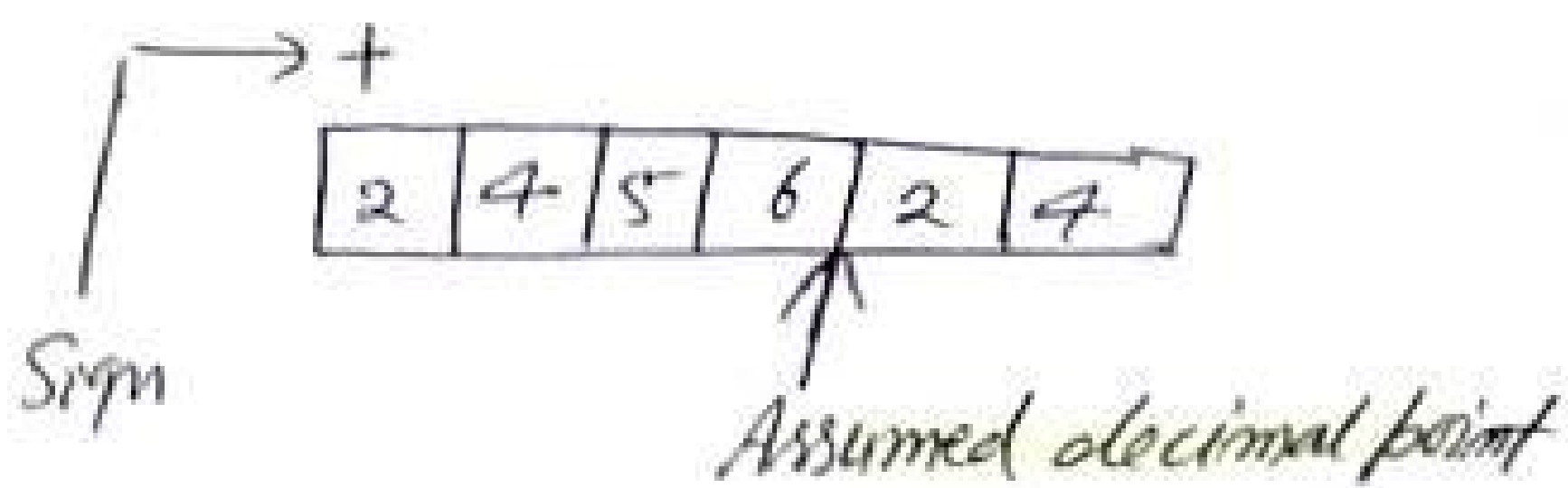
Integer arithmetic deals with integer operands i.e. numbers without fractional parts. It is used mainly in counting and as subscripts.

Real arithmetic uses numbers with fractional parts as operands and is used in most computations.

Due to economic considerations computers are designed such that each location (also called a word) in memory stores only a finite number of digits. Consequently all operands in arithmetic operations have only a finite number of digits.

For illustrative purpose we will assume that a (hypothetical) Computer has a memory in which each location can store 6 digits and has provision to store one or more signs. One method of representing real numbers in such a computer could be to assume a fixed position for the decimal point and stores all numbers (after appropriate shifting if necessary)

with an assumed decimal point as shown in following figure -

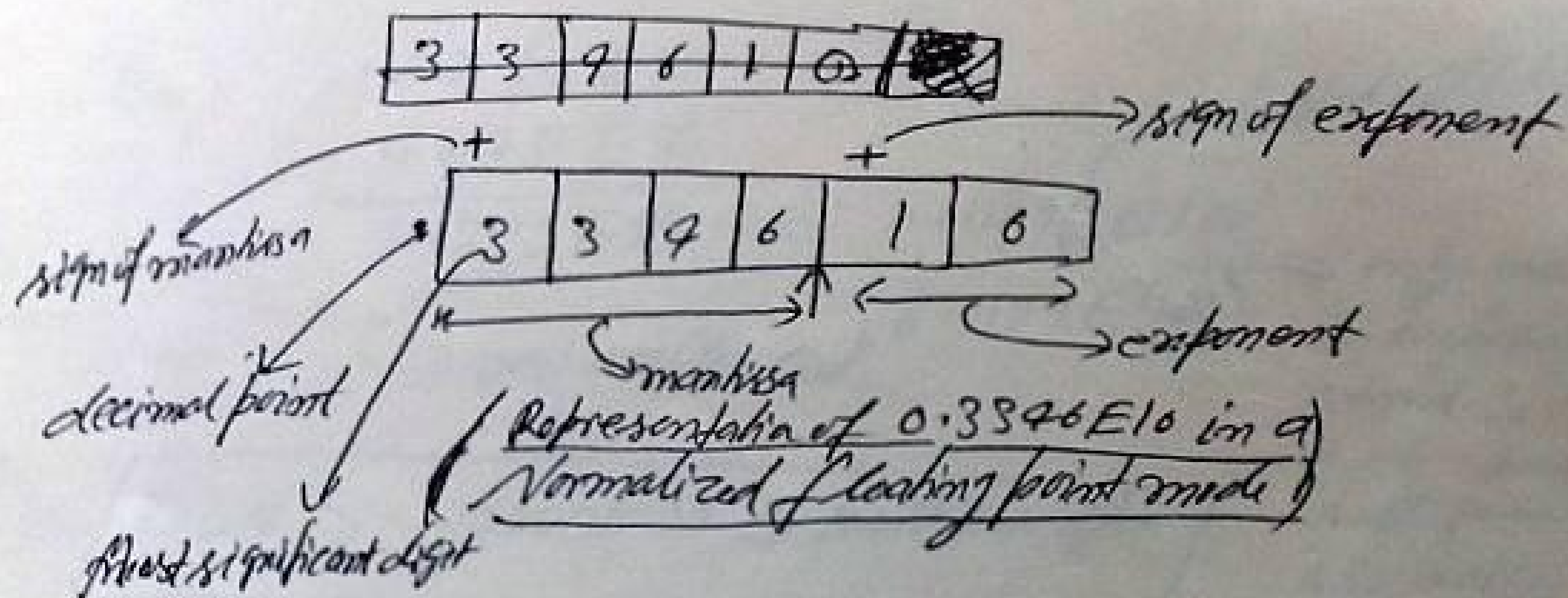


(A memory location ~~storing~~ storing the number 2456.24)

In such a convention is used the maximum & minimum (in magnitude) numbers that may be stored are 9999.99 & 0000.01 respectively. This range is quite inadequate in practice and a different convention for representing real numbers is adopted. This convention aims to preserve the maximum number of significant digits in a real number and also increase the range of values of real numbers stored. This representation is called the normalized floating point mode of representing and storing real numbers. In this mode a real number is expressed as a combination of a mantissa and an exponent. The mantissa is made less than 1 and greater than or equal to 0.1 and the exponent is the power of 10 which multiplies the mantissa. ~~It is~~

$0.1 \leq \text{mantissa} < 1$ → in a normalized floating point mode

e.g. 33.46×10^8 i.e. $0.3346 \times 10^{10} \Rightarrow 0.3346 E 10$
 (mantissa) (exponent)



✓ Normalization:- Shifting of the mantissa to the left till its most significant digit is non-zero is called normalization.

Thus, using this notation we can store $.9999 \times 10^{99}$ to $.1000 \times 10^{-99}$ which is larger than that used in the fixed decimal point notation.

Arithmetic Operations with Normalized Floating Point Numbers -

✓ ① Addition - If two numbers represented in normalized floating point notation are to be added then the exponents of the two numbers must be made equal and the mantissa shifted appropriately.
e.g. Add $.4546E5$ to $.5433E5$

$$\Rightarrow .9979E5$$

② Add $.4546E5$ to $.5433E7$ → chopped off (discarded)

Solⁿ $.4546E5 = .004546E7$

$$\begin{array}{r} .004546E7 \\ + .5433E7 \\ \hline .547846E7 \end{array}$$

③ $\begin{array}{r} .6434E3 \\ + .4845E3 \\ \hline 1.1279E3 \end{array} \Rightarrow .011279E4$ → chopped off (discarded)
 $\Rightarrow 0.1127E4$

④ $\begin{array}{r} .6434E99 \\ + .4845E99 \\ \hline 1.1279E99 \end{array} \Rightarrow 0.11279E100$ → chopped off (discarded)
→ new overflow cond. $0.1127E(100)$ → only maximum 2 digit in exponent could be stored. So it could not be stored and error message will come and overflow will arise.

Result $> .9999E99 \Rightarrow$ leads to overflow

Q. What do you mean by Overflow & Underflow in context of Arithmetic operations with Normalized floating point numbers.

✓ Subtraction — ① Subtract $.1432E-4$ from $.5452E-3$

(10)

$$\begin{array}{r} \text{i.e. } .5452E-3 \\ - .1432E-4 \\ \hline \end{array} \Rightarrow \begin{array}{r} .5452E-3 \\ - .01432E-3 \\ \hline .4509E-3 \text{ Ans} \end{array} \rightarrow \text{chopped off (i.e. discarded)}$$

$$\begin{array}{r} \text{② } .5452E3 \\ - .5424E3 \\ \hline .0028E3 \Rightarrow .2800E1 \text{ Ans.} \end{array}$$

$$\begin{array}{r} \text{③ } .5452E-99 \\ - .5424E-99 \\ \hline .0028E-99 \Rightarrow .2800E-101 \rightarrow \text{underflow cond}^n \end{array}$$

i.e. Result $< .1000E-99 \Rightarrow \text{Underflow}$

✓ Multiplication — Multiply mantissa and add exponent and express result in normalized floating point mode (i.e. mantissa is normalized)

✓ Division — Mantissa of ~~number~~ numerator is divided by the ^{mantissa of} denominator and the denominator's exponent is subtracted from the numerator's exponent.

Q1. Round off the following numbers to four significant digits and then calculate absolute error, relative error and percentage error.

(i) 75462 (ii) 537.261 (iii) 21.46235

Solⁿ (i) Here, the number is 75462 i.e. $n = 75462$

After rounded off to four significant digits, number is 75460 i.e. $n_1 = 75460$.

Now, absolute error, $e_a = |n - n_1| = |75462 - 75460| = 2$.

$$\therefore e_a = 2$$

$$\text{relative error, } e_r = \left| \frac{e_a}{n} \right| = \left| \frac{2}{75462} \right| = 2.65 \times 10^{-5}$$

$$\therefore e_r = 2.65 \times 10^{-5}$$

$$\text{percentage error, } e_p = e_r \times 100 = 2.65 \times 10^{-5} \times 100 = 2.65 \times 10^{-3}$$

$$\therefore e_p = 2.65 \times 10^{-3}$$

(ii) Here, the number is 537.261 i.e. $n = 537.261$

After rounded off to four significant digits, the number is 537.3 i.e. $n_1 = 537.3$

Now, absolute error, $e_a = |n - n_1| = |537.261 - 537.3| = 0.039$

$$\therefore e_a = 0.039$$

$$\text{relative error, } e_r = \left| \frac{e_a}{n} \right| = \left| \frac{0.039}{537.261} \right| = 7.259 \times 10^{-5}$$

$$\therefore e_r = 7.259 \times 10^{-5}$$

$$\text{percentage error, } e_p = e_r \times 100 = 7.259 \times 10^{-5} \times 100 = 7.259 \times 10^{-3}$$

$$\therefore e_p = 7.259 \times 10^{-3}$$

(iii) Here, the number is 21.46235 i.e. $n = 21.46235$

After rounded off to four significant digits, the number is 21.46 i.e. $n_1 = 21.46$

~~Now, after rounded~~

Now, absolute error, $e_a = |n - n_1| = |21.46235 - 21.46| = 0.00235 = 2.35 \times 10^{-3}$

$$\therefore e_a = 2.35 \times 10^{-3}$$

$$\text{relative error, } e_r = \left| \frac{e_a}{n} \right| = \left| \frac{2.35 \times 10^{-3}}{21.46235} \right| = 0.000109494 = 1.09 \times 10^{-4}$$

$$\therefore e_r = 1.09 \times 10^{-4}$$

$$\text{percentage error, } e_p = e_r \times 100 = 1.09 \times 10^{-4} \times 100 = 1.09 \times 10^{-2}$$

$$\therefore e_p = 1.09 \times 10^{-2}$$

Q2. If 0.333 is the approximate value of $\frac{1}{3}$, find absolute, relative and percentage errors.

Solⁿ Here, we have true value $x_t = \frac{1}{3}$ and approximate value $x_a = 0.333$.

Now, the absolute error, $e_a = |x_t - x_a| = \left| \frac{1}{3} - 0.333 \right| = \left| \frac{1}{3} - \frac{333}{1000} \right| = 0.00033$

$$\therefore e_a = 0.00033$$

$$\text{relative error, } e_r = \left| \frac{e_a}{x_t} \right| = \left| \frac{0.00033}{\frac{1}{3}} \right| = 0.00099$$

$$e_r = 0.00099$$

$$\text{percentage error, } e_p = e_r \times 100 = 0.00099 \times 100 = 0.099$$

$$e_p = 0.099$$

Q3. If $x = 0.005998$ then find the relative error if x is truncated to three decimal digits.

Solⁿ Here, $x_t = 0.5998 \times 10^{-2}$ (in normalized floating point form)

$\therefore x_a = 0.599 \times 10^{-2}$ (after truncating to three decimal places)

$$\therefore \text{Relative error, } e_r = \left| \frac{x_t - x_a}{x_t} \right| = \left| \frac{0.5998 \times 10^{-2} - 0.599 \times 10^{-2}}{0.5998 \times 10^{-2}} \right| = 0.133 \times 10^{-2}$$

$$\therefore e_r = 0.133 \times 10^{-2}$$

Q4. The solution of the a problem is given as 4.536. If it is known that absolute error in the solution is less than 0.01 then find the interval within which the exact value must lie.

Solⁿ Approximate value of the solution is given as $x_a = 4.536$.

If x be the exact value of the solution, then according to the question,

$$\text{we have } |x - x_a| < 0.01 \text{ i.e. } |x - 4.536| < 0.01$$

Now, from the definition of absolute value, we have

$$-0.01 < x - 4.536 < 0.01$$

$$\text{or } (4.536 - 0.01) < x < (4.536 + 0.01)$$

$$\text{or } 4.526 < x < 4.546$$

Hence, the value of the solution must lie in the interval (4.526, 4.546).

Q5. An approximate value of π is given by 3.1428571 and its true value is 3.1415926. Find absolute and relative errors.

Sol.ⁿ Here, $x_t = 3.1415926$ and $x_a = 3.1428571$.

$$\therefore \text{absolute error, } e_a = |x_t - x_a| = |3.1415926 - 3.1428571| = 0.0012645$$

$$\therefore e_a = 0.0012645$$

$$\text{and, relative error, } e_r = \left| \frac{e_a}{x_t} \right| = \left| \frac{0.0012645}{3.1415926} \right| = 0.000462$$

$$\therefore e_r = 0.000462$$

Q6. Use the series $\log_e \left(\frac{1+x}{1-x} \right) = 2 \left(x + \frac{x^3}{3} + \frac{x^5}{5} + \dots \right)$ to compute the value of $\log_e(1.2)$ correct to seven decimal places and find the number of terms retained.

Sol.ⁿ Let $\frac{1+x}{1-x} = 1.2 \Rightarrow x = \frac{1}{11}$

If we retain n terms of the series then

$$(n+1)^{\text{th}} \text{ term} = 2 \cdot \frac{x^{2n+1}}{2n+1} = 2 \cdot \frac{\left(\frac{1}{11}\right)^{2n+1}}{2n+1} = \frac{2}{(2n+1) \cdot 11^{2n+1}}$$

$\therefore$ For seven decimal accuracy,

$$\frac{2}{(2n+1) \cdot 11^{2n+1}} < \frac{1}{2} \times 10^{-7}$$

$$\text{or } 4 \times 10^7 < (2n+1) \cdot 11^{2n+1}$$

$$\text{or } (2n+1) \cdot 11^{2n+1} > 4 \times 10^7$$

$$\Rightarrow n \geq 3$$

Hence, retaining the first three terms of the given series, we get

$$\log_e(1.2) = 2 \left(x + \frac{x^3}{3} + \frac{x^5}{5} \right) \text{ at } x = \frac{1}{11}$$

$$\text{or } \log_e(1.2) = 0.1823215$$